

Pharmacology of Cholinomimetics: Direct-Acting Muscarinic and Nicotinic Receptor Agonists

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Make.
Heal.
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I am available to groups and individuals for pharmacology help and discussions by appointment.

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Following completion of the preparation materials, students should be able to:

1. PK properties:

Compare the class pharmacokinetic properties of the direct-acting muscarinic and nicotinic cholinomimetics to those of the individual agents, including the tendency to cross the blood-brain barrier and the effect of pH on excretion in urine.

2. Physiology-Pharmacology relationship → Mechanism of action and effects:

Relate the ANS neurotransmitter chemistry, receptor characteristics, and functional integration to the structure-activity relationship, mechanism of action, and effects of each of the direct-acting cholinomimetics.

3. PK properties + Mechanism of action + Effect → Mechanisms of Drug interactions

Apply the pharmacokinetic properties, therapeutic effects, adverse effects, and potential drug-disease and drug-drug interactions of the individual agents to the selection of drug therapy for the individual patient.

4. PK + Effects + Drug-disease and Drug-drug interactions → Cautious use:

Predict the cautions and contraindications for use based on mechanisms of action, adverse effects, and drug interactions.

5. Poisoning:

Correlate the signs and symptoms of nicotinic toxicity with the mechanisms of action and the effects of the drugs.

Preparation Materials

Required

- **ScholarRx Bricks | Practice Questions**

Suggested supplemental resources if you wish to use them:

- Lecture Video | Dr. Goldstein's Word handout | Guided Reading Questions
- Links to a textbook and pharmacology review books are provided on the next slide.

SUGGESTIONS:

- *Use the resources that work best for you.*
- *You do not need to study all of them. (ScholarRx Bricks & Practice Questions are required.)*
- *Work through the GUIDED READING QUESTIONS with pen/pencil and paper.*

Try answering them in your OWN WORDS first without looking at the readings, and then again after reviewing.

- *Practice questions (not graded): Simple Recall and Case Vignettes*

Links to resources

ScholarRx Bricks Bricks: General Pharmacology > Autonomic Drugs

Autonomic Drugs: Introduction to Autonomic Drugs > Foundations and Frameworks <https://exchange.scholarrx.com/brick/autonomic-drugs-foundations-and-frameworks>

Drugs That Alter Parasympathetic Tone > Parasympathomimetics > Cholinomimetics <https://exchange.scholarrx.com/brick/cholinomimetic-agents>

Access Medicine Katzung's Basic & Clinical Pharmacology, 16e, 2024

Chapter 6: Introduction to Autonomic Pharmacology

<https://nyit.idm.oclc.org/login?url=https://accessmedicine.mhmedical.com/content.aspx?bookid=3382§ionid=281747427>

Chapter 7: Cholinoceptor-Activating & Cholinesterase-Inhibiting Drugs

<https://nyit.idm.oclc.org/login?url=https://accessmedicine.mhmedical.com/content.aspx?bookid=3382§ionid=281747500>

NOTE: THE REVIEW BOOKS LISTED BELOW HAVE PRACTICE QUESTIONS.

LWW Health Library, Medical Education: Lippincott's Illustrated Reviews: Pharmacology; 8e, 2023

Chapter 3: The Autonomic Nervous System

<https://nyit.idm.oclc.org/login?url=https://premiumbasicsciences.lwwhealthlibrary.com/content.aspx?sectionid=253325317&bookid=3222>

Chapter 4: Cholinergic Agonists

<https://nyit.idm.oclc.org/login?url=https://premiumbasicsciences.lwwhealthlibrary.com/content.aspx?sectionid=253325401&bookid=3222>

Access Medicine Katzung's Pharmacology: Examination & Board Review, 14e, 2024; Chapter 6: Introduction to Autonomic Pharmacology

<https://nyit.idm.oclc.org/login?url=https://accessmedicine.mhmedical.com/content.aspx?bookid=3461§ionid=285589591>

Chapter 7: Cholinoceptor-Activating & Cholinesterase-Inhibiting Drugs

<https://nyit.idm.oclc.org/login?url=https://accessmedicine.mhmedical.com/content.aspx?bookid=3461§ionid=285589820>

Autonomic Nervous System Overview



The autonomic nervous system is like a spider web
—intricate, interconnected, and interdependent—

The following key points are intended help the student grasp the main ideas and concepts by organizing and summarizing the “need to know” information. Key points may also serve as a concise reference to facilitate review.

Hints for studying the pharmacology of the ANS

1. The core concepts of ANS pharmacology are built on knowledge of the **PHYSIOLOGIC** actions and responses of the autonomic nervous system.

Knowledge of the of the actions of acetylcholine on its various nicotinic and muscarinic receptors and the cellular responses to cholinergic stimulation are essential for understanding the effects of the drugs that mimic or block the cholinergic effects at the ganglia, the adrenal medulla, and the effector organs.

An overview is given in various resources and on pages 1-6 of the notes handout.

2. **Start from the Big Picture:** WHY are they used? HOW will they achieve the desired therapeutic outcomes? WHAT is the common thread that connects all of them?

3. **CLASS** actions and effects form the foundation. All of the drugs will relate to the class actions.

4. When you are comfortable with your knowledge of the above, it's time to learn the drug names and the few drug-specific effects (usually pharmacokinetic properties) that differ from the class effects.

Key points: Definitions

Acetylcholine (ACh)	Neurotransmitter of the ANS acting at the ganglia and effector organs of the parasympathetic nervous system (PNS) and sweat glands
Acetylcholinesterase (AChE)	Enzyme that degrades acetylcholine in the synapse
Norepinephrine (NE)	Neurotransmitter of the ANS at the effector organs of the sympathetic nervous system (SNS), also secreted by the adrenal medulla
Epinephrine (Epi)	Hormone of the ANS secreted by the adrenal medulla and acting at the effector organs of the SNS
Ionotropic receptors	Membrane-bound receptor proteins that respond to ligand binding by opening an ion channel and allowing ions to flow into the cell, either increasing or decreasing the likelihood that an action potential will fire
G protein-coupled receptors (GPCRs)	Membrane-bound receptor proteins that, when activated by a ligand, activate enzymes (second messengers), operate ion channels, and initiate intracellular cascades
Metabotropic receptors	GPCRs that indirectly link to ion channels through a signaling mechanism involving second messengers

Key points: Acetylcholine

- All preganglionic efferent autonomic nerve fibers and somatic motor fibers to skeletal muscle, all parasympathetic postganglionic fibers and sympathetic fibers innervating sweat gland release the neurotransmitter **acetylcholine** – these are **cholinergic neurons**.
- Acetylcholine is synthesized in the cytosol from acetyl-CoA and choline and stored in vesicles. It is released into the synapse when an action potential reaches the nerve terminal and triggers sufficient influx of calcium ions via N-type calcium channels.
- Acetylcholine is degraded in the synaptic cleft by acetylcholinesterase.
- Acetylcholine activates nicotinic and muscarinic receptors in the CNS and periphery:
 - preganglionic nicotinic receptors (ionotropic receptors)
 - neuromuscular nicotinic receptors (ionotropic receptors)
 - muscarinic G protein-coupled receptors (metabotropic receptors) on effector organs

Key points: Acetylcholine receptors and actions of cholinomimetic drugs

- Nicotinic receptors (nAChRs) are ligand-gated nonselective cation channels permeable to Na^+ and K^+ ; some are permeable to Ca^{2+} . Two molecules of ACh binding at the interface between the α subunits and the adjacent subunits open the nicotinic receptor ion channel, permitting one or more specific ions to diffuse through.
- Five subgroups of muscarinic receptors, mChRs, (M_{1-5}) have been described. They are:
 - M_1 , M_3 , M_5 couple to the $G_{q/11}$ PCR family of G proteins.
 - M_2 , M_4 preferentially couple to the $G_{i/o}$ PCR family of G proteins.
- Cholinomimetic drugs – acetylcholine-like drugs – have numerous and complex actions.
- A drug's structure determines its relative selectivity for receptor type – muscarinic or nicotinic receptors – and resistance to hydrolysis – **the structure-activity relationship**.

Key points: Properties of direct-acting cholinomimetics

- The direct-acting cholinomimetics are choline esters and naturally occurring alkaloids that bind specific receptor types and trigger an action potential or activate a second messenger signaling system of that receptor subtype. That is, they are nicotinic or muscarinic receptor agonists.
- The choline esters, which are quaternary ammonium compounds, are poorly absorbed from the GI tract and have limited ability to cross the blood-brain barrier. They are short-acting and rapidly eliminated by the kidneys.
- The alkaloids, such as nicotine and pilocarpine, are weak bases and tertiary amines. They are readily absorbed and cross the blood-brain barrier. They are metabolized by the liver. The metabolites are primarily eliminated by the kidney.
- The various agents are relatively nonselective and differ in their spectrums of action – that is, the degree of muscarinic versus nicotinic stimulation and in their relative affinities for the muscarinic receptor subtypes.

Key points: Direct-acting cholinomimetics' clinical uses and adverse effects

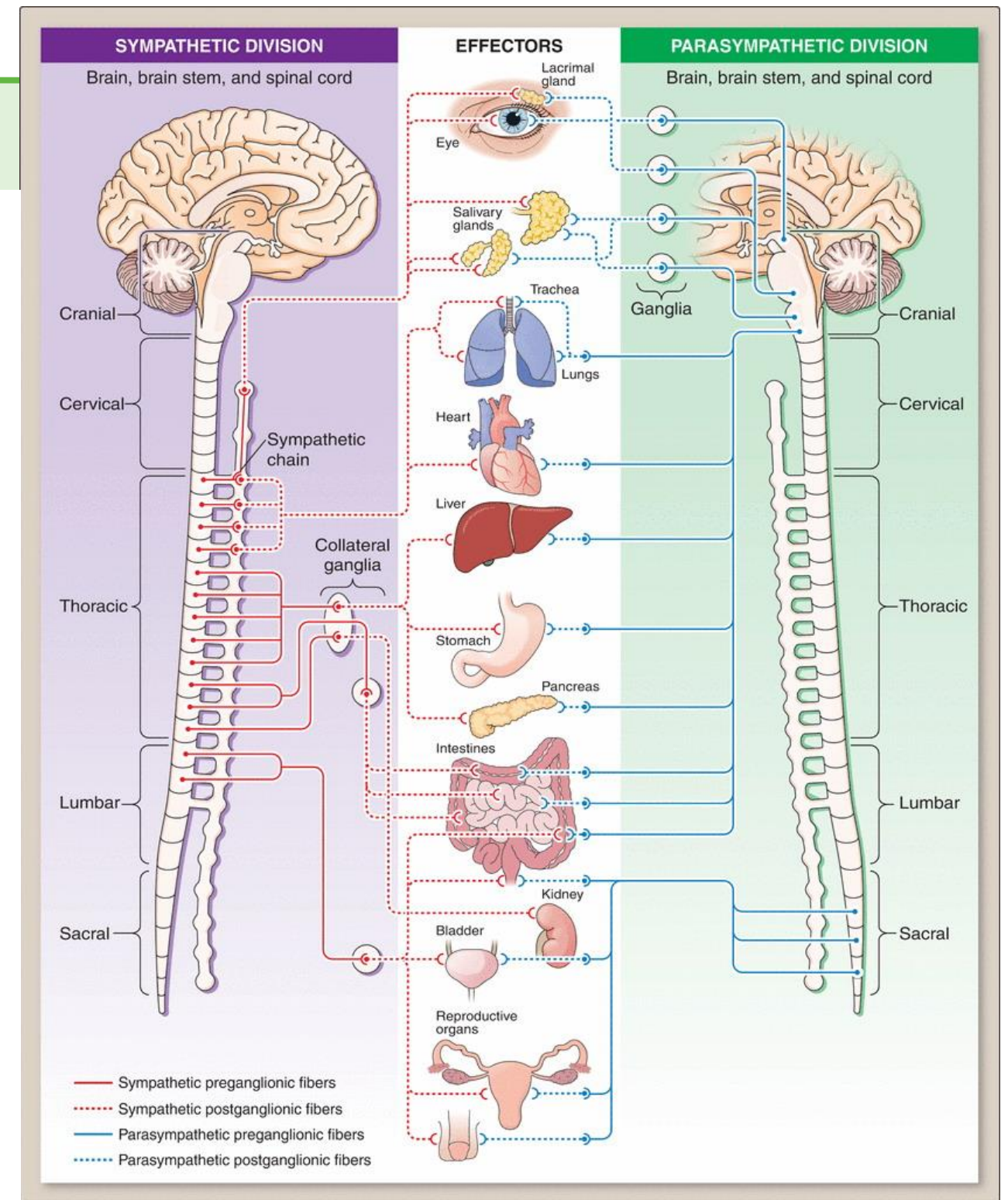
- The various direct-acting cholinomimetics are used in the treatment of urinary bladder disorders, xerostomia, glaucoma, miotic agents, and in the diagnosis of bronchial hyperreactivity.
- Muscarinic toxicity is characterized by CNS stimulation (uncommon with choline esters, which do not cross BBB), miosis, ciliary muscle spasm (spasm of accommodation), bronchoconstriction, excessive gastrointestinal and genitourinary smooth muscle activity, increased secretory activity (sweat glands, airway, gastrointestinal tract, lacrimal glands), and vasodilation.
- Nicotinic receptor agonists are smoking cessation aids.
- Nicotine's toxic effects include ganglionic stimulation and depolarizing blockade and neuromuscular end plate depolarization leading to fasciculations and then flacid paralysis. CNS toxicity includes stimulation (including convulsions) followed by depression and paralysis of respiratory muscles.
- Varencline is a synthetic neuronal $\alpha_4\beta_2$ nAChR partial agonist that decreases nicotine cravings and withdrawal symptoms. It is well absorbed and minimally metabolized with excellent bioavailability.

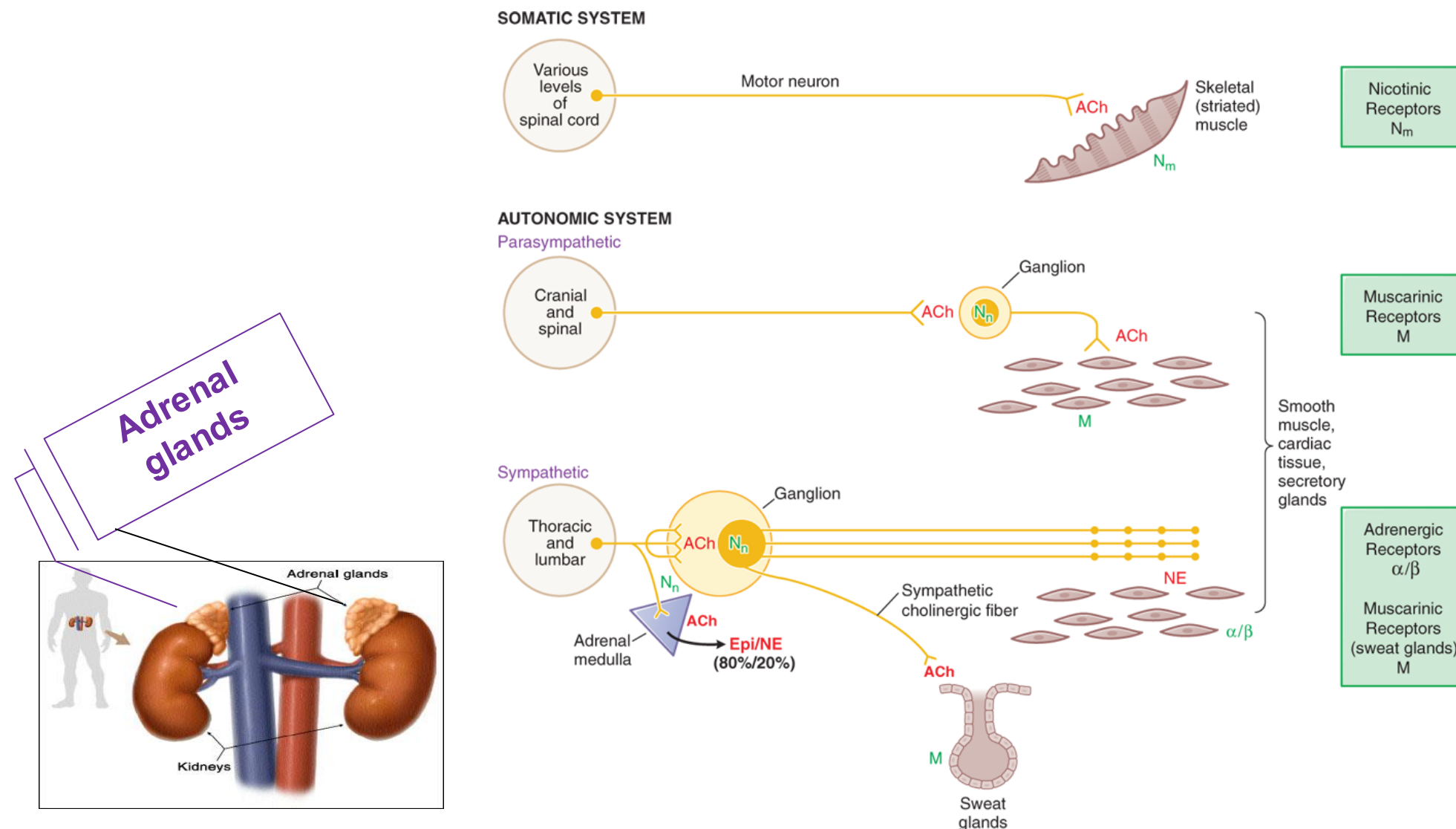
- Indirect-acting cholinomimetics will be described in another lecture.
- The sympathetic nervous system and sympathomimetics will be described in another lecture.

Overview of the Autonomic Nervous System Functions

From: **3 The Autonomic Nervous System**, Figure 3.3
Lippincott® Illustrated Reviews: Pharmacology, 8e, 2023

- The ANS regulates respiration, circulation, digestion, body temperature, metabolism, sweating, and the secretions of certain endocrine glands.
 - **Parasympathetic craniosacral division of the ANS:**
 - ↓HR; ↓BP; constricts pupil (protects retina); stimulates bronchoconstriction and mucus secretion in the lungs; stimulates GI tract; empties urinary bladder/rectum
 - **Sympathetic thoracolumbar division of the ANS:**
 - Constitutively active
 - “Fight or flight” response → ↑HR; ↑BP; blood flow shifts from skin/splanchnic to skeletal muscles; ↑blood glucose; bronchioles dilate; pupils dilate; CNS facilitation of response
 - **Enteric NS:** self-contained (autonomous); GI functions
- Cholinergic neurons stimulate smooth muscle contractions and secretions. Parasympathetic input and vaso-vagal reflexes are crucial for regulating numerous gut functions. Sympathetic regulation is inhibitory for maintaining homeostasis during stress or emergency situations.





Source: Laurence L. Brunton, Björn C. Knollmann:
Goodman & Gilman's: The Pharmacological Basis of Therapeutics, 14e:
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Comparative features of somatic motor nerves and efferent nerves of the autonomic nervous system. The principal neurotransmitters, ACh and NE, are shown in red. The receptors for these transmitters, nicotinic (N) and muscarinic (M) cholinergic receptors, α and β adrenergic receptors, are shown in green. Somatic nerves innervate skeletal muscle directly at a specialized synaptic junction, the motor end plate, where ACh activates N_m receptors. Autonomic nerves innervate smooth muscles, cardiac tissue, and glands. Both parasympathetic and sympathetic systems have ganglia, where ACh is released by the preganglionic fibers; ACh acts on N_n receptors on the postganglionic nerves. ACh is also the neurotransmitter at cells of the adrenal medulla, where it acts on N_n receptors to cause release of EPI and NE into the circulation. ACh is the dominant neurotransmitter released by postganglionic parasympathetic nerves and acts on muscarinic receptors. The ganglia in the parasympathetic system are near or within the organ being innervated, with generally a one-to-one relationship between pre- and postganglionic fibers. NE is the principal neurotransmitter of postganglionic sympathetic nerves, acting on α or β adrenergic receptors. Autonomic nerves form a diffuse pattern with multiple synaptic sites. In the sympathetic system, the ganglia are generally far from the effector cells (e.g., within the sympathetic chain ganglia). Preganglionic sympathetic fibers may make contact with a large number of postganglionic fibers.

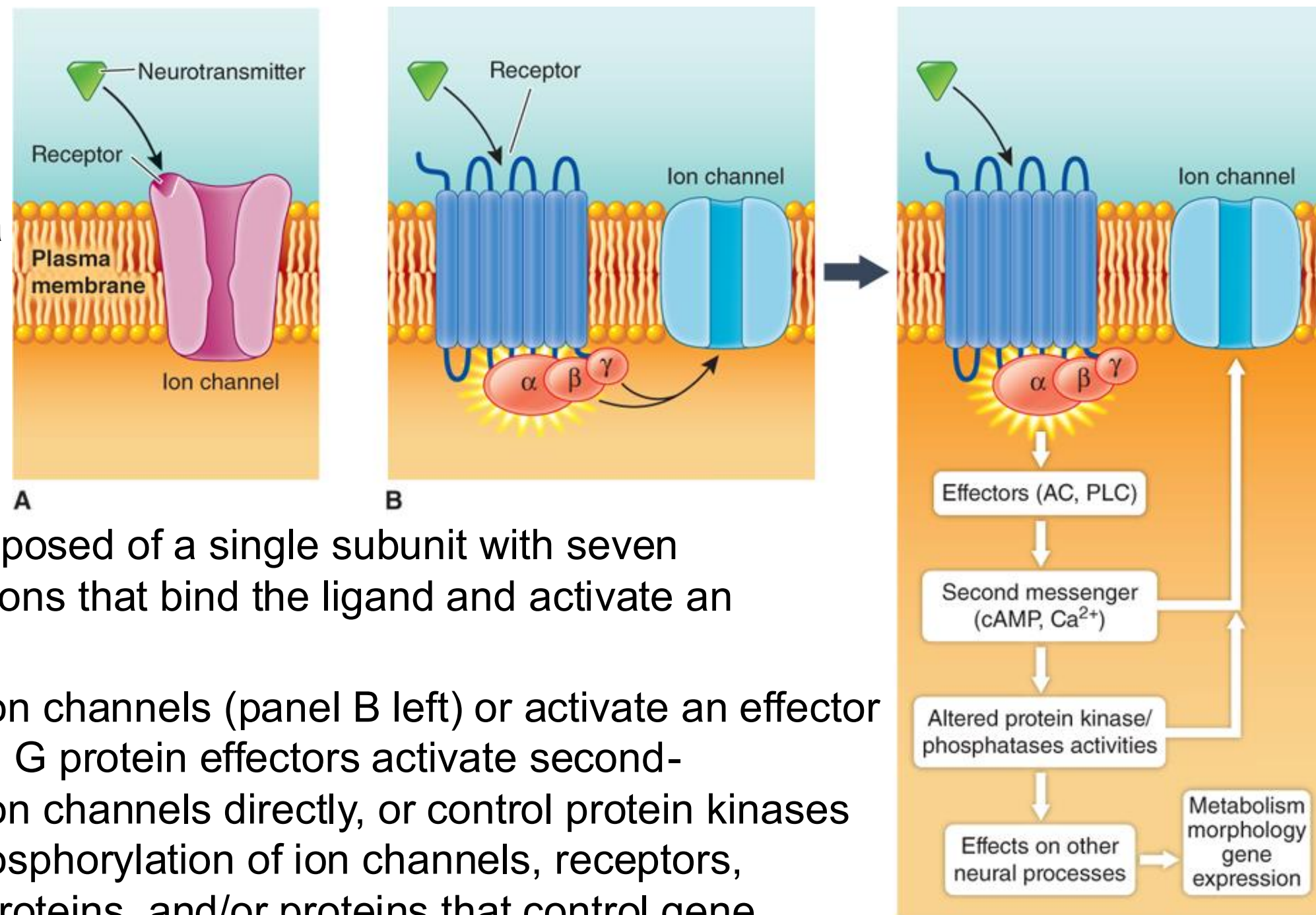
Two types of neurotransmitter receptors.

A. Ionotropic receptors are ligand-gated channels that contain both a neurotransmitter (NT) binding domain and an integral ion channel domain. Most ionotropic receptors are composed of four or five subunits.

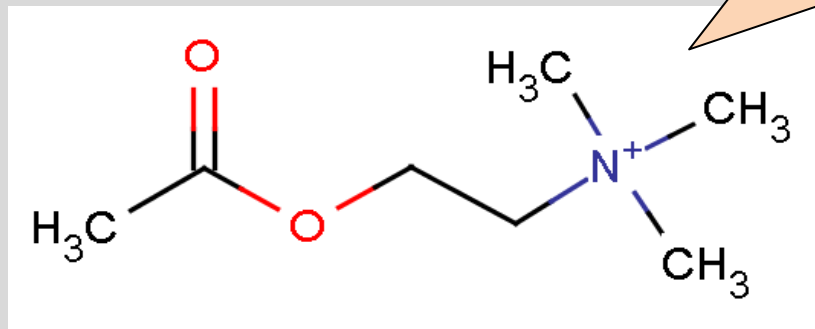
B. Metabotropic receptors are composed of a single subunit with seven membrane-spanning α -helical regions that bind the ligand and activate an associated G protein.

G proteins can directly modulate ion channels (panel B left) or activate an effector (panel B right) such as AC or PLC. G protein effectors activate second-messengers, which can regulate ion channels directly, or control protein kinases and phosphatases that modulate phosphorylation of ion channels, receptors, metabolic enzymes, cytoskeletal proteins, and/or proteins that control gene expression.

AC, adenylyl cyclase; cAMP, cyclic adenosine monophosphate; PLC, phospholipase C.

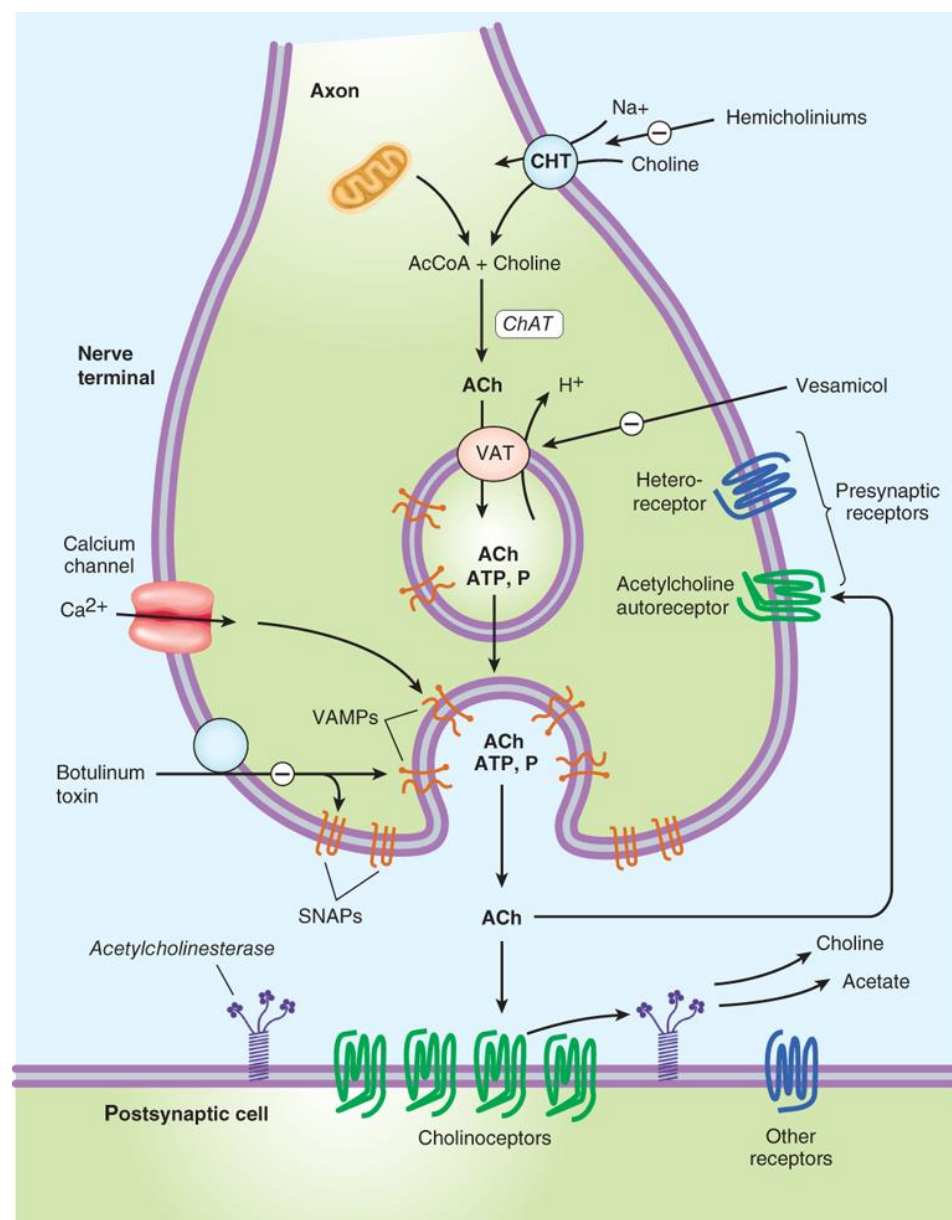


quaternary
ammonium
compound



ACETYLCHOLINE

synthesis, storage, release, degradation



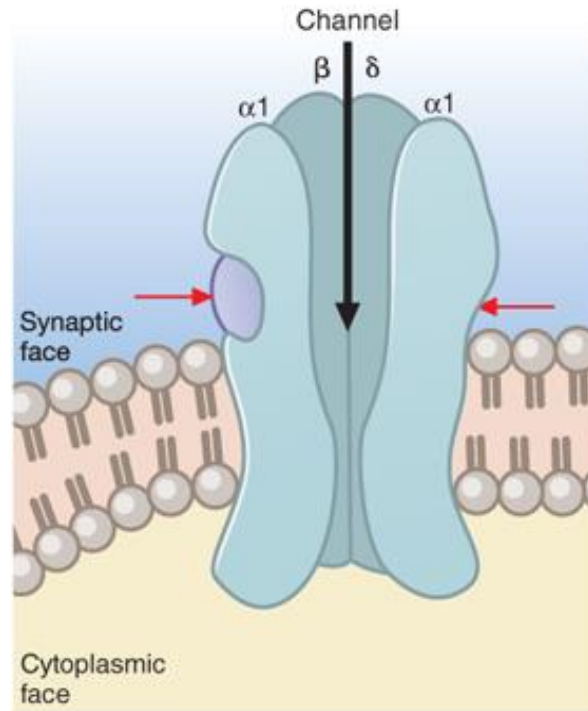
Source: Bertram G. Katzung, Todd W. Vanderah:
Basic & Clinical Pharmacology, Fifteenth Edition
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1. Choline transport
2. $\text{Ch} + \text{AcetylCoA} = \text{ACh}$
3. VAT $\rightarrow$ Transport of ACh into storage vesicle
4. Axon terminal depolarization $\rightarrow \text{Ca}^{2+}$ influx
5. Vesicle attachment and fusion
6. Exocytosis of ACh & cotransmitters into junctional cleft
7. $\text{AChE} \rightarrow \text{Ch} + \text{acetate} \rightarrow \text{terminate action}$
8. Choline recycled; acetate diffuses away

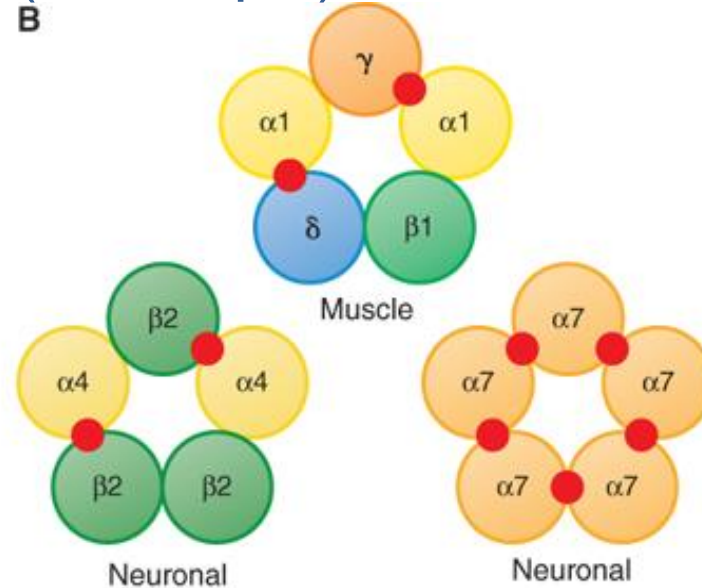
Schematic illustration of a generalized cholinergic junction (not to scale). Choline is transported into the presynaptic nerve terminal by a sodium-dependent choline transporter (CHT). This transporter can be inhibited by hemicholinium drugs. In the cytoplasm, acetylcholine is synthesized from choline and acetyl-CoA (AcCoA) by the enzyme choline acetyltransferase (ChAT). Acetylcholine (ACh) is then transported into the storage vesicle by a vesicle-associated transporter (VAT), which can be inhibited by vesamicol. Peptides (P), adenosine triphosphate (ATP), and proteoglycan are also stored in the vesicle. Release of transmitters occurs when voltage-sensitive calcium channels in the terminal membrane are opened, allowing an influx of calcium. The resulting increase in intracellular calcium causes fusion of vesicles with the surface membrane and exocytotic expulsion of acetylcholine and cotransmitters into the junctional cleft (see text). This step can be blocked by botulinum toxin. Acetylcholine's action is terminated by metabolism by the enzyme acetylcholinesterase. Receptors on the presynaptic nerve ending modulate transmitter release. SNAPs, synaptosomal nerve-associated proteins; VAMPs, vesicle-associated membrane proteins.

nicotinic ACh receptors are
ion channels (ionotropic)

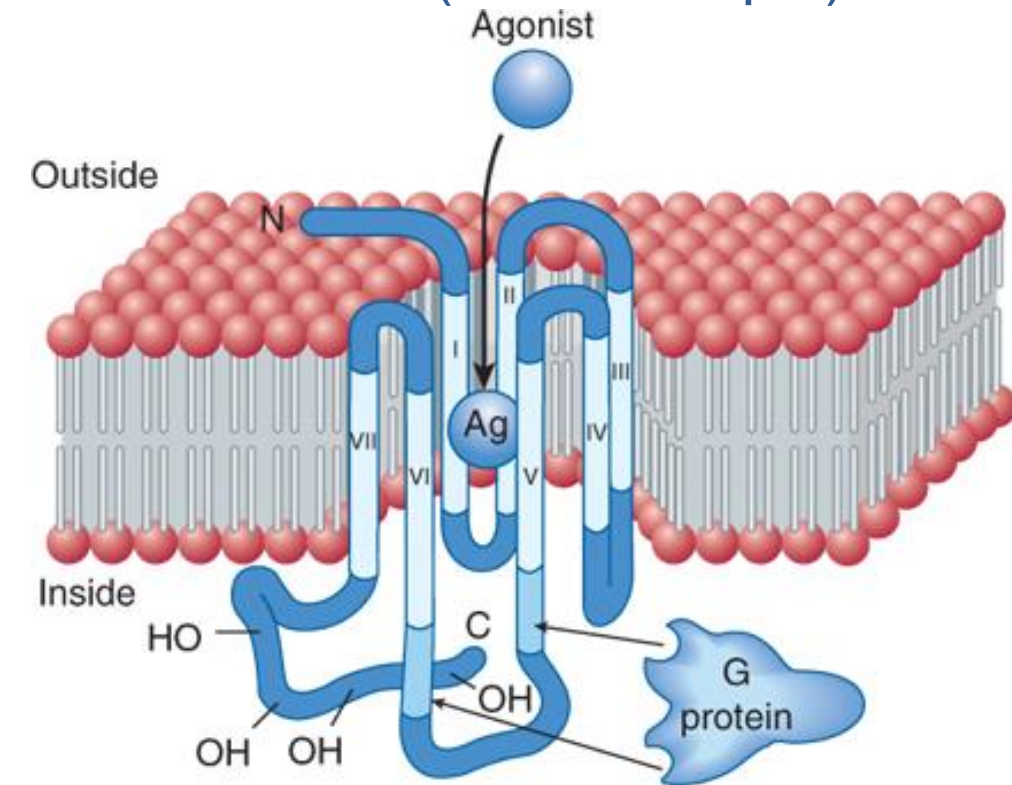
A



B



muscarinic ACh receptors are
GPCRs (metabotropic)



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Goodman & Gilman's: The Pharmacological Basis of Therapeutics, 14e:
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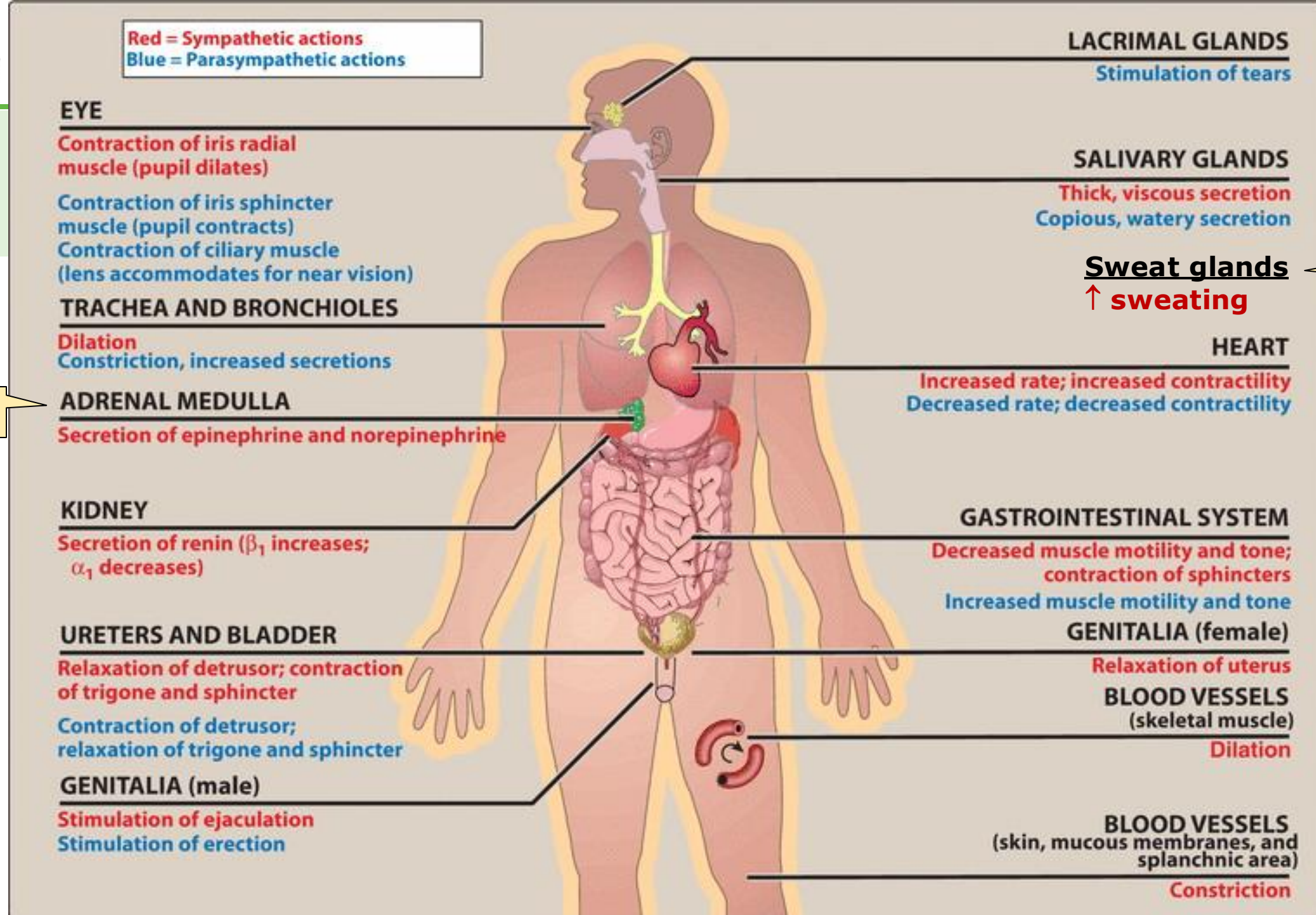
Simultaneous binding of 2 ACh molecules at the interface between the α subunits and the adjacent subunits significantly increases the channel opening probability, permitting one or more specific ions to diffuse through.

N_N and N_M nicotinic Rs - ionotropic

M_1, M_3, M_5 $G_{q/11}$ PCRs $\rightarrow$ PLC $\rightarrow$ IP_3 & DAG $\rightarrow$
 $\uparrow Ca^{2+} + \uparrow PKC$, respectively

M_2, M_4 $G_{i/o}$ $\rightarrow$ inhibits adenylyl cyclase $\rightarrow$ $\downarrow cAMP$
 $\rightarrow \uparrow K^+$ conductance and $\downarrow Ca^{2+}$ conductance

Responses of effector organs to cholinergic stimulation



N_N receptor

M_3 receptor

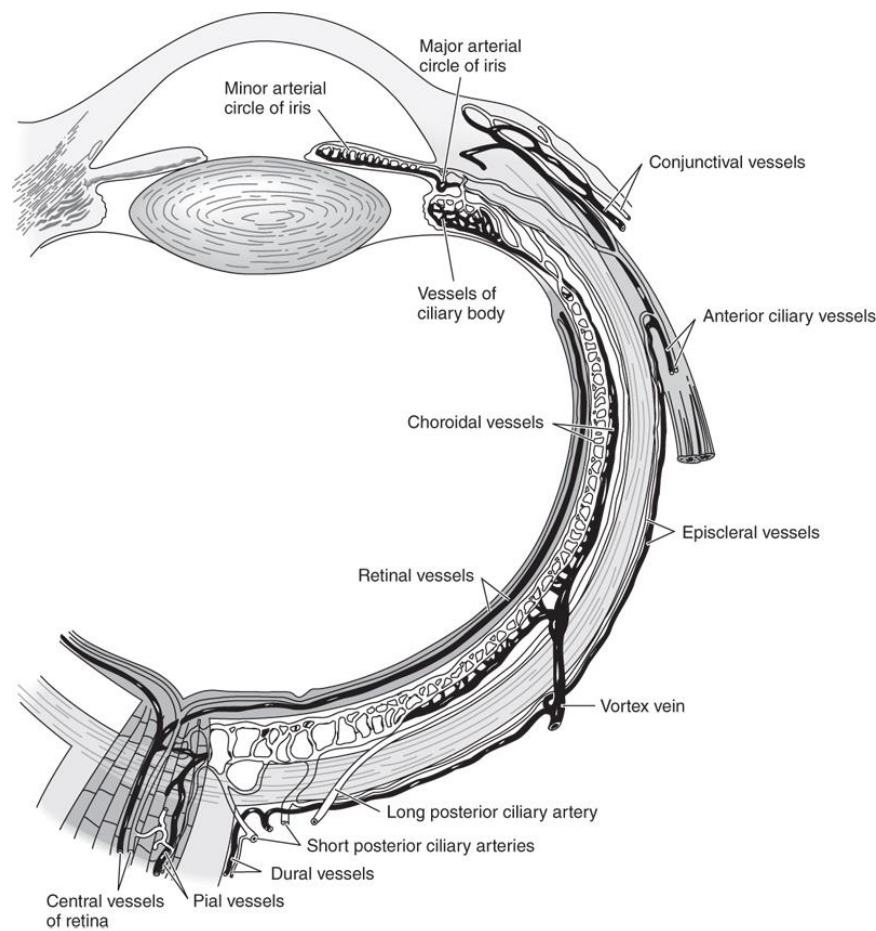
Nonneuronal receptors are also expressed on immune cells, vasculature (β_2 , M_3), and bronchi (β_2)

Lippincott® Illustrated Reviews: Pharmacology, 8e, 2023

From: **3 The Autonomic Nervous System**, Figure 3.4

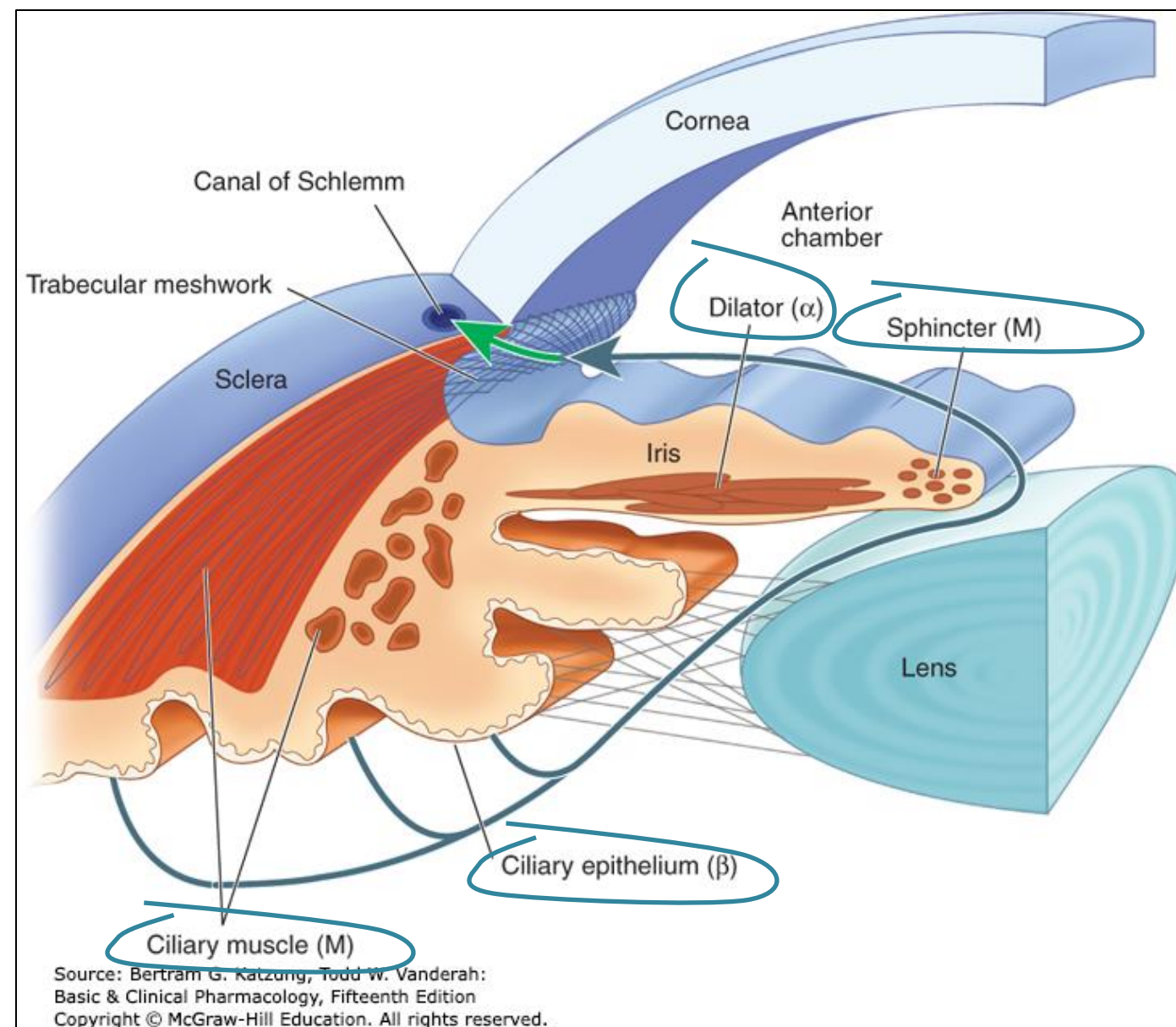
Actions of sympathetic and parasympathetic nervous systems on effector organs. Copyright © Wolters Kluwer

Slide adapted by LG



Source: P. Riordan-Eva, J. J. Augsburger: Vaughan & Asbury's General Ophthalmology, Nineteenth Edition Copyright © McGraw-Hill Education. All rights reserved.

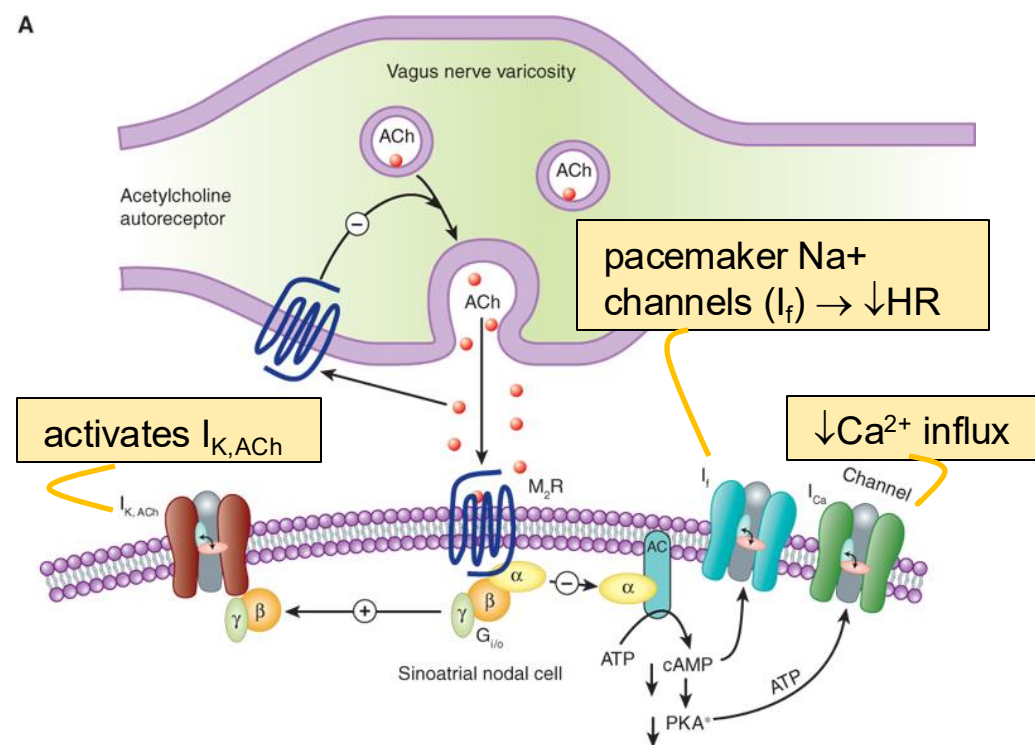
McGraw-Hill Chapter 1 Anatomy & Embryology of the Eye, Riordan-Eva P, Augsburger JJ. *Vaughan & Asbury's General Ophthalmology*, 19e; 2017. Figure 1-4



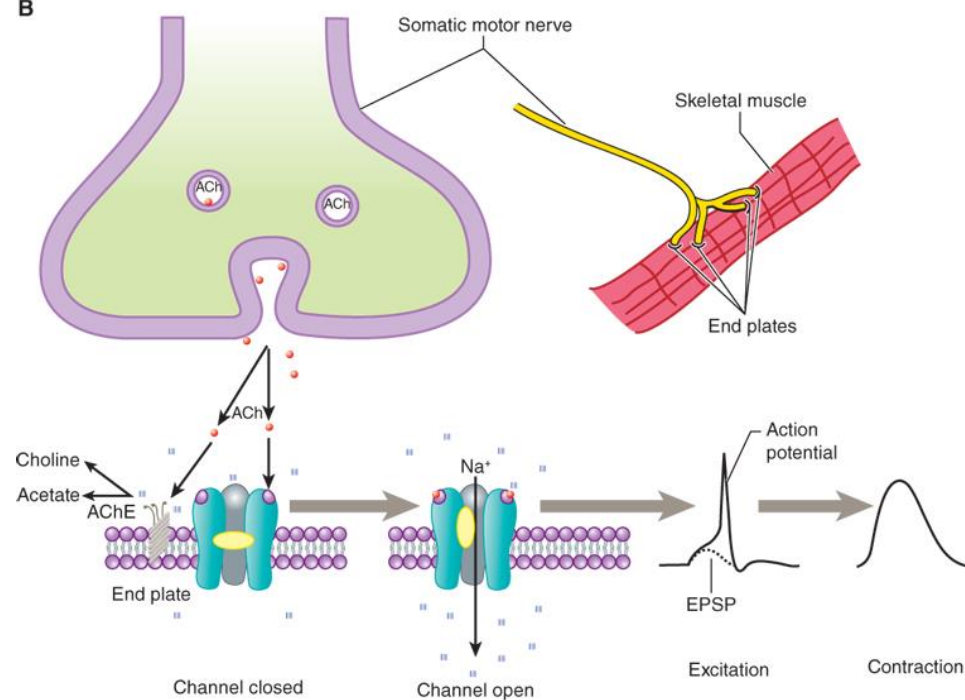
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Structures of the anterior chamber of the eye. Tissues with significant autonomic functions and the associated ANS receptors (α , β , M) are shown in this schematic diagram. Aqueous humor is secreted by the epithelium of the ciliary body, flows into the space in front of the iris, flows through the trabecular meshwork, and exits via the canal of Schlemm (arrow). Blockade of the β adrenoceptors associated with the ciliary epithelium causes decreased secretion of aqueous. Blood vessels (not shown) in the sclera are also under autonomic control and influence aqueous drainage.

A



B



Source: Todd W. Vanderah:
Basic & Clinical Pharmacology, Sixteenth Edition
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Citation: Chapter 7 Cholinoceptor-Activating & Cholinesterase-Inhibiting Drugs, Vanderah TW. *Katzung's Basic & Clinical Pharmacology*, 16th Edition; 2024, Figure 7-4. Available at: <https://accessmedicine.mhmedical.com/content.aspx?bookid=3382§ionid=281747500>
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Muscarinic signaling in the heart / Nicotinic signaling in skeletal muscle

Panel A:

Muscarinic transmission to the sinoatrial node in the heart **slows heart rate**.

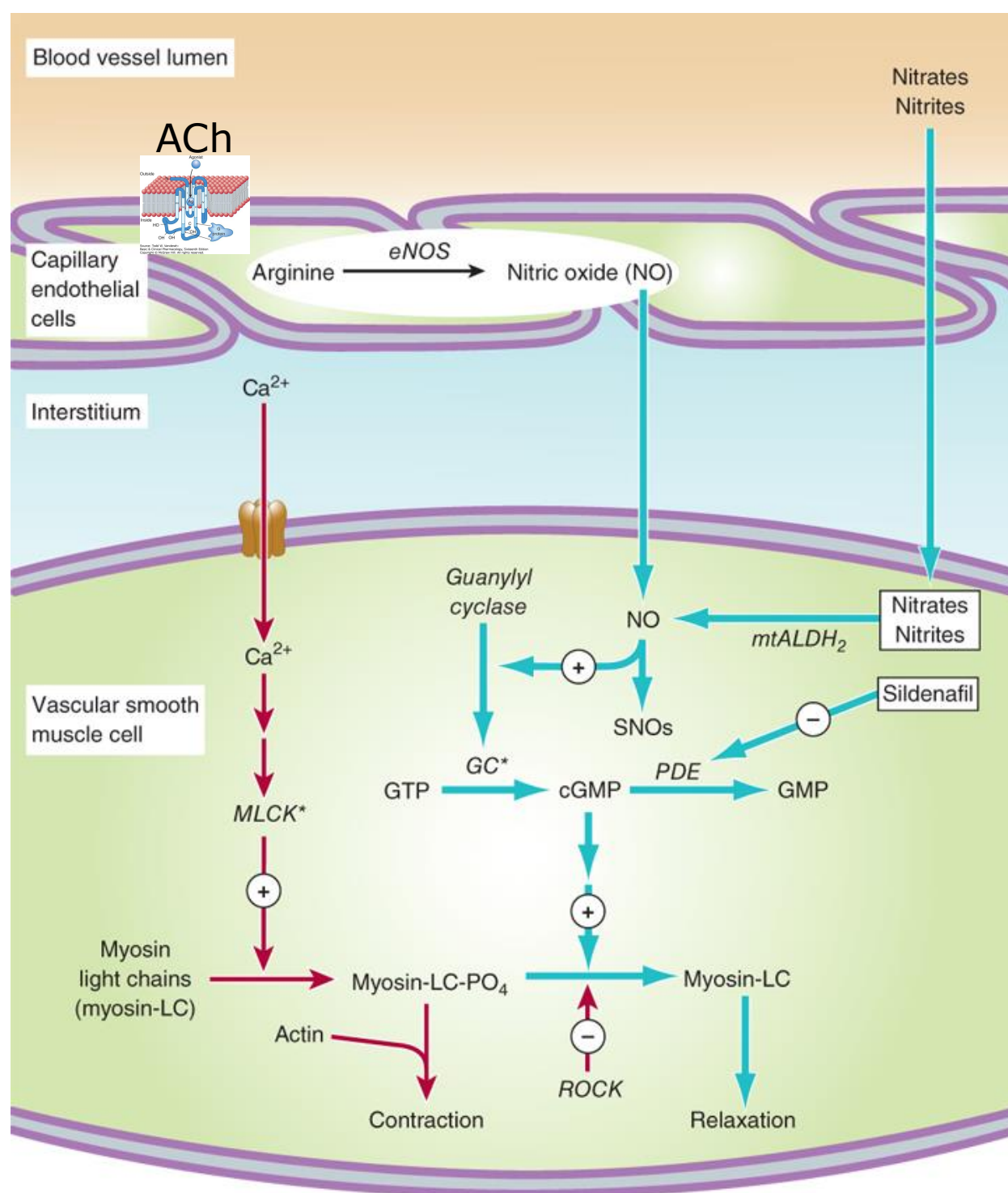
- Acetylcholine (ACh) released from a varicosity of a postganglionic cholinergic axon interacts with a sinoatrial node cell muscarinic receptor (M_2R) linked via $G_{i/o}$ to K^+ channel opening, which causes hyperpolarization, and inhibition of cAMP synthesis.
- Reduced cAMP shifts the voltage-dependent opening of **pacemaker channels (I_f)** to more negative potentials and reduces the phosphorylation and availability of **L-type Ca^{2+} channels (I_{Ca})**.
- Released ACh also acts on an axonal muscarinic autoreceptor to cause inhibition of ACh release (autoinhibition).
- Muscarinic transmission also slows conduction through the AV node and reduces the force of contraction. (These effects are not shown in the figure.)

Panel B:

Nicotinic transmission at the skeletal neuromuscular junction creates an end plate potential (if strong enough).

- ACh released from the motor nerve terminal interacts with subunits of the pentameric nicotinic receptor to open it, allowing Na^+ influx to produce an excitatory postsynaptic potential (EPSP).
- The EPSP depolarizes the muscle membrane, generating an action potential, and triggering contraction.
- Acetylcholinesterase (AChE) in the extracellular matrix hydrolyzes (inactivates) ACh

Slide adapted by LG



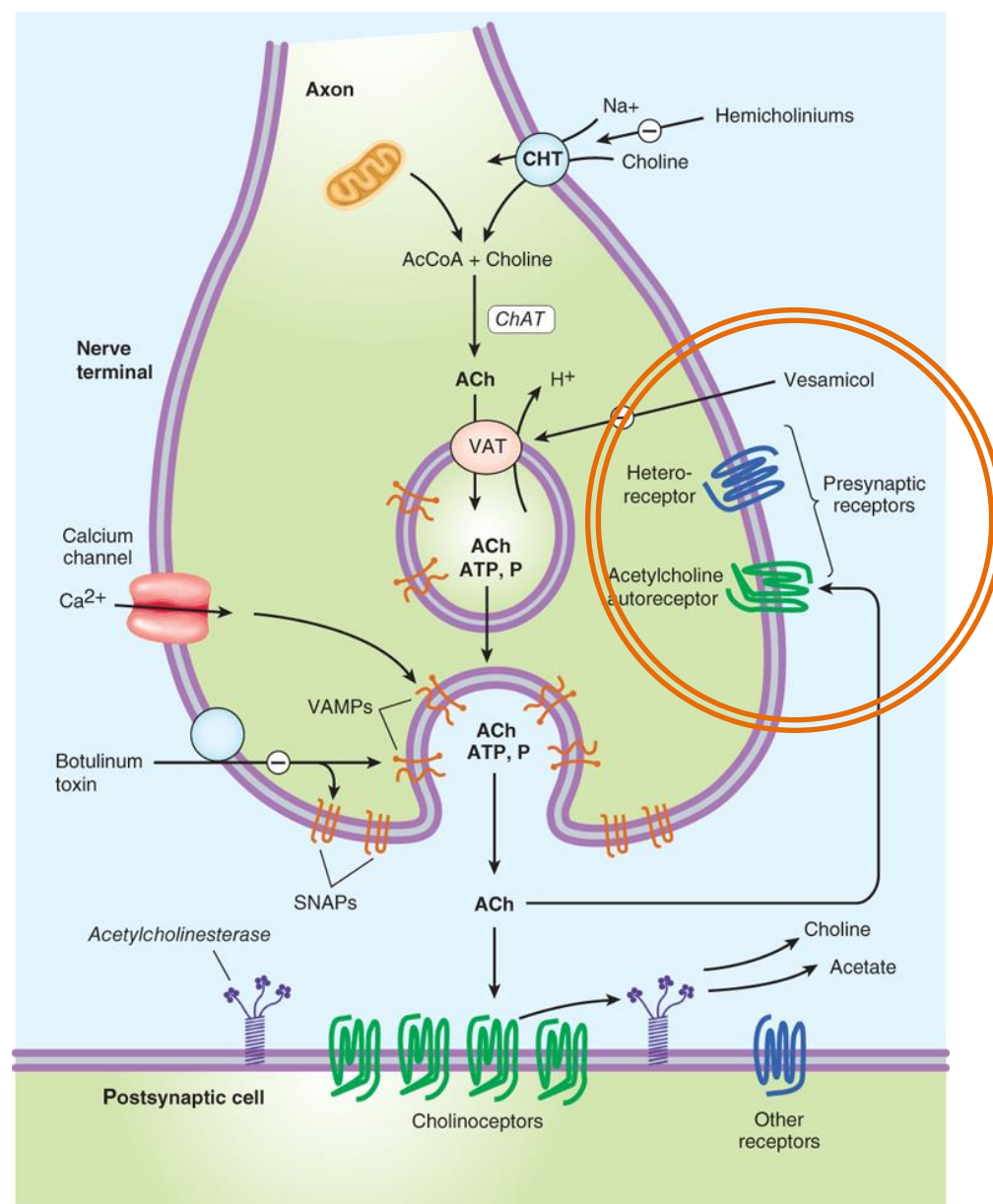
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Mechanism of nitric oxide (NO)-mediated vasodilation

- Endothelial nitric oxide synthase (eNOS) is activated by fluid sheer stress, numerous agonists, interactions with various substrates, cofactors, regulatory proteins, and more.
- NO is released, diffuses into vascular smooth muscle cells, and induces relaxation.

Effect of acetylcholine intravenous infusion (investigative use – not used in clinical practice)

- ACh IV infusion activates nonneuronal M₃ receptors on vascular endothelium, which stimulates eNOS and NO synthesis.
- Small IV doses cause vasodilation with a reduction in blood pressure, often accompanied by a reflex increase in heart rate
- (ACh action is not specifically shown on the figure.)



Source: Bertram G. Katzung, Todd W. Vanderah:
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- **Local integration:** Feedback by modulatory autoreceptors and heteroreceptors on presynaptic nerve membranes regulate neurotransmitter release.

- **Systemic reflexes:**

Homeostatic mechanisms regulate blood pressure, gastrointestinal motility, bladder tone, airway smooth muscle, and other processes.

For example, a drop in blood pressure is detected by the baroreceptors in the carotid arteries and aortic arch → signals the sympathetic nervous system in the CNS → increases heart rate and force of contraction → heart pumps more blood + vasoconstriction → increase blood pressure

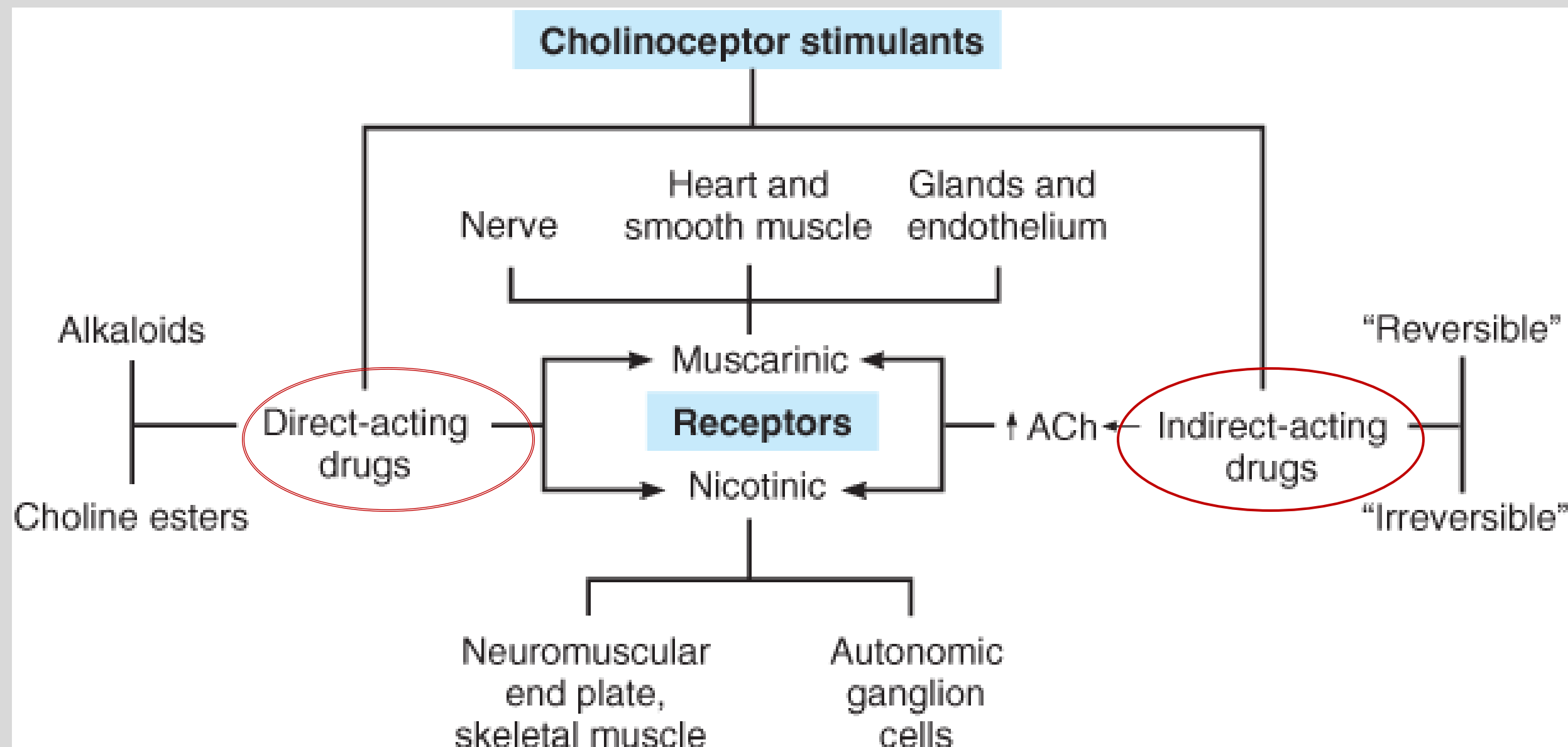
- ❖ Remember, the nervous system is an interconnected, intricate network that ensures the body maintains homeostasis and appropriate responses to various stimuli.

Schematic illustration of a generalized cholinergic junction (not to scale). Choline is transported into the presynaptic nerve terminal by a sodium-dependent choline transporter (CHT). This transporter can be inhibited by hemicholinium drugs. In the cytoplasm, acetylcholine is synthesized from choline and acetyl-CoA (AcCoA) by the enzyme choline acetyltransferase (ChAT). Acetylcholine (ACh) is then transported into the storage vesicle by a vesicle-associated transporter (VAT), which can be inhibited by vesamicol. Peptides (P), adenosine triphosphate (ATP), and proteoglycan are also stored in the vesicle. Release of transmitters occurs when voltage-sensitive calcium channels in the terminal membrane are opened, allowing an influx of calcium. The resulting increase in intracellular calcium causes fusion of vesicles with the surface membrane and exocytotic expulsion of acetylcholine and cotransmitters into the junctional cleft (see text). This step can be blocked by botulinum toxin. Acetylcholine's action is terminated by metabolism by the enzyme acetylcholinesterase. Receptors on the presynaptic nerve ending modulate transmitter release. SNAPs, synaptosomal nerve-associated proteins; VAMPs, vesicle-associated membrane proteins.

Check your knowledge

- What is the definition of cholinergic receptors?
-
- What are the cholinergic receptor subtypes?
-
- Where in the body are cholinergic receptors located?
-

Pharmacology of Cholinomimetics



Direct-acting Muscarinic Receptor Agonists

Choline esters and Tertiary amines

Choline Esters

- *Acetylcholine
- *Methacholine
- *Carbachol
- *Bethanechol

Tertiary Amines

- *Pilocarpine
- *Cevimeline



Muscarine is an alkaloid of *Amanita muscaria* and other mushrooms, native throughout temperate regions of the Northern hemisphere.
https://commons.wikimedia.org/wiki/File:Amanita_muscaria.jpg#/media/File:Amanita_muscaria.jpg



Pilocarpine is an alkaloid of *Pilocarpus*, a native plant of the neotropics of South America.

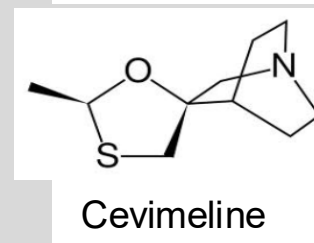
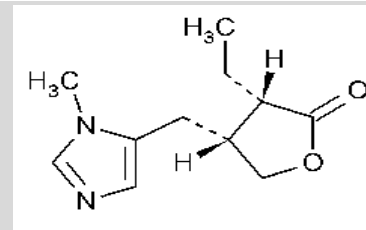
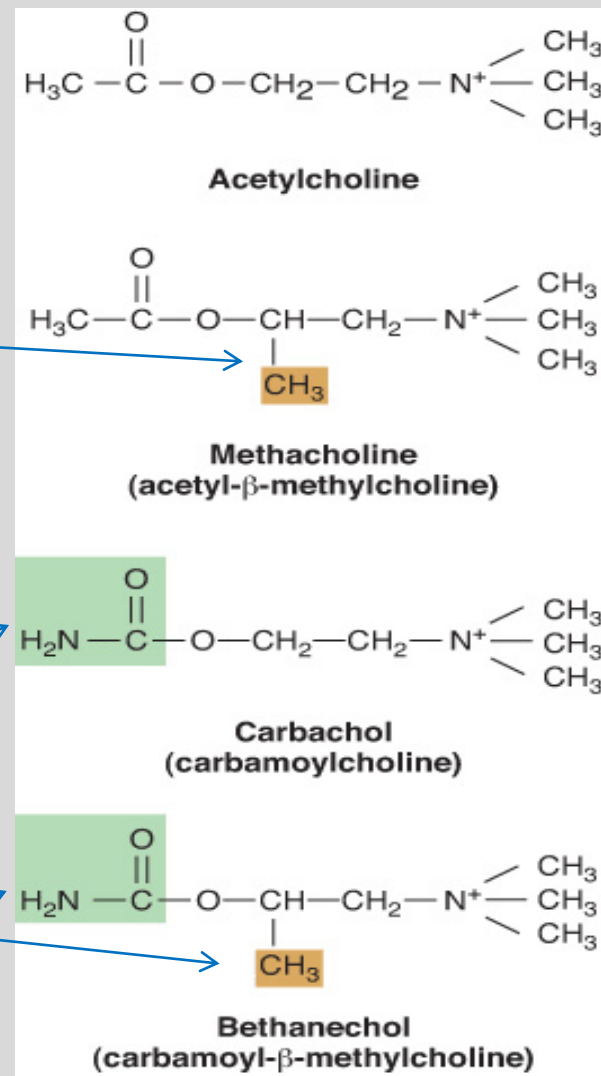
Choline Esters quaternary ammonium compounds

↑resistance to ChE hydrolysis +
↓ nicotinic activity

Resistance to ChE hydrolysis.
Carbachol lacks β-CH₃ so has
muscarinic + nicotinic activity

Resistance to ChE hydrolysis
+ ↓ nicotinic activity

Natural Alkaloid and Derivative
tertiary amines
(orally absorbed)



Muscarinic Agonist	Susceptibility to Cholinesterase	Nicotinic Activity
Acetylcholine	+++	++
Methacholine	More resistant to cholinesterase hydrolysis. Longer duration of action.	+
Carbachol	Even more resistant to cholinesterase hydrolysis. Even longer duration of action.	+++
Bethanechol		-
Pilocarpine	-	-
Muscarine	-	-

Class Pharmacokinetics

Choline esters:

Hydrophilic (charged compounds)

Poorly absorbed

Poorly distributed into the CNS $\Rightarrow$ do not readily cross membrane barriers

Rapidly eliminated by kidney

Only bethanechol is administered orally for systemic effects.

Tertiary amines

Pilocarpine:

Well absorbed, crosses blood-brain barrier

Plasma/tissue esterases and hepatic metabolism

Excreted in urine

Cevimeline (synthetic):

Rapid absorption, limited CNS penetration

Hepatic metabolism, excreted in urine

Therapeutic uses of muscarinic agonists by acting on M3 receptors

Acetylcholine	Instillation, anterior chamber: Induction of miosis for ophthalmologic surgery	
Methacholine	Inhalation: Bronchoprovocation diagnostic challenge test for bronchial airway hyper-reactivity in patients suspected of having asthma ➤ Rationale: Airway hyperresponsiveness may be the sole evidence of airway dysfunction/asthma. Duration 30-45 minutes unless reversed by rapid-acting bronchodilator Reversal agent: β2-adrenergic agonist, i.e., albuterol (reversal occurs in ~5 min)	
Carbachol	Topical eye drops: Glaucoma; induction of miosis during ophthalmologic surgery	
Bethanechol	Oral: Urinary retention due to neurogenic bladder acute post-operative or post-partum urinary retention	
Pilocarpine	Topical eye drops: Induction of miosis Acute angle-closure glaucoma Chronic open-angle glaucoma	Oral: Sialagogue Symptomatic treatment of dry mouth (xerostomia) caused by Sjögren's syndrome or salivary gland hypofunction resulting from radiotherapy
Cevimeline	Oral: Sialagogue for the symptomatic treatment of dry mouth (xerostomia) caused by Sjögren's syndrome or salivary gland hypofunction resulting from radiotherapy.	

Adverse effects of muscarinic agonists by acting on M3 receptors

(not a complete list)

All mAChR agonists	Systemic effects: bradycardia, hypotension with reflex tachycardia, dyspnea ➤ Tachycardia increases cardiac oxygen demand and increases blood pressure. Caution: recent MI, uncontrolled hypertension, aortic aneurysm, urinary obstruction, GI obstruction, any condition that could be adversely affected by a cholinergic agent (respiratory disease; peptic ulcer disease; hyperthyroidism; GI, biliary, urinary obstruction)	
Acetylcholine	Ophthalmic: cloudy vision, corneal edema	
Methacholine	Inhalation: bronchospasm (can be severe), Caution: Baseline FEV1 <60%	
Carbachol	Ophthalmic: blurred vision, corneal edema, headache	
Bethanechol	Oral: Blood pressure decrease with tachycardia, bronchoconstriction, diaphoresis, lacrimation (tearing), salivation, nausea, abdominal cramps, urinary urgency,	
Pilocarpine	Topical eye drops: blurred vision, impaired vision, accommodation disturbance	Oral: dizziness, headache, asthenia (feeling of weakness) Peripheral cholinergic effects (like bethanechol)
Cevimeline	Oral: CNS and peripheral cholinergic effects typical of the class	

Check your knowledge

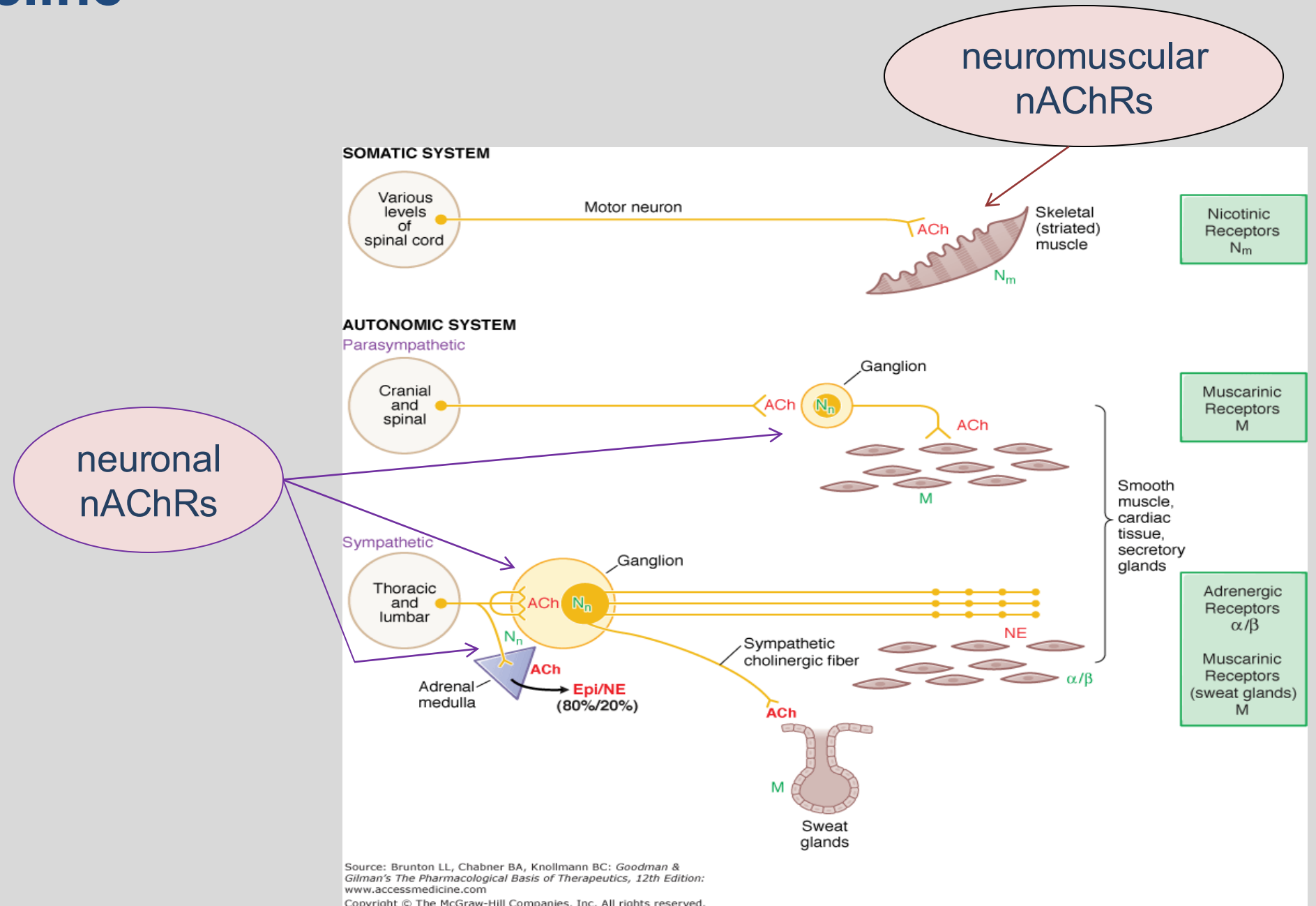
- Acetylcholinesterase can hydrolyze 5000 acetylcholine molecules per second. Why, then, is the duration of action of ACh instilled in the anterior chamber as long as 6 hours?
-
- An asymptomatic 6-year-old patient suspected of having asthma has a positive bronchoprovocation test using an orally inhaled cholinergic agent. What are the mechanism of action and effects of the administered drug?
-
- What are the mechanism-related side effects of muscarinic receptor agonists?
-

Direct-Acting Nicotinic Receptor Agonists

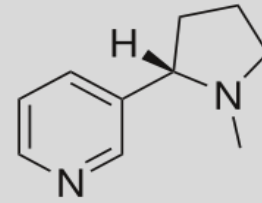
Nicotine and Varenicline



Nicotine is an alkaloid of *Nicotiana glauca*, a plant of the nightshade family indigenous to the Americas and other areas.
Painting by William Rossin posters



Nicotine Pharmacokinetics



Nicotine

A	Well absorbed from buccal mucosa, respiratory tract, skin Formulations: buccal, nasal spray, transdermal patch, vaping liquid
D	Distributes to CNS and peripheral tissues
M	Liver (major), kidney, lung Metabolites: Cotinine (major); >20 metabolites
E	Renal (primarily metabolites): pH-dependent - pH 7: ~2% unchanged nicotine in urine (~98% reabsorbed) - pH <5: ~23% unchanged nicotine in urine (~77% reabsorbed) (weak base in more acidic medium → greater fraction in BH ⁺ species – ion trapping)
t_{1/2}	Inhalation and nasal 1-2 h; transdermal ~4 h Cotinine (biomarker) ~16 h

Nicotine has **biphasic effects**.



(1) Stimulation: Small dose of agonist → rapid depolarization of postsynaptic membranes → may facilitate impulse transmission

(2) Depolarizing blockade: Results from prolonged agonist occupancy

Although membrane voltage returns to resting level with continued presence of the nicotinic agonist, the receptor becomes **desensitized** to agonist and **refractory** to reversal, which results in:



**Persistent depression of
all autonomic ganglia and neuromuscular junctions**

CNS:

Stimulation of prejunctional nicotinic receptors → *neurotransmitter release*

Stimulation and pleasure-reward:

Nicotine influences release of other transmitters dopamine, norepinephrine, acetylcholine, other biogenic amines, excitatory amino acids

Other acute CNS effects (dose-dependent):

Dizziness, tremor, respiratory excitation, vomiting

Large doses → convulsions (CNS) and respiratory failure (N_M block), death

Chronic exposure to nicotine: ↑ density of CNS nAChRs

Nicotine's peripheral effects:

Stimulation of postjunctional nicotinic receptors → *excitatory synaptic transmission and pre- to postsynaptic signal transmission*

Sympathetic Stimulation

Direct ganglionic actions cause:

- vasoconstriction
- ↑ heart rate
- ↑ blood pressure



baroreceptor stimulation



reflexive responses

Adrenal Medulla

- ↑ catecholamine release

GI

- ↑ tone and motor activity of the bowel

Exocrine glands

- initially, ↑ salivary and bronchial secretions
- followed by inhibition

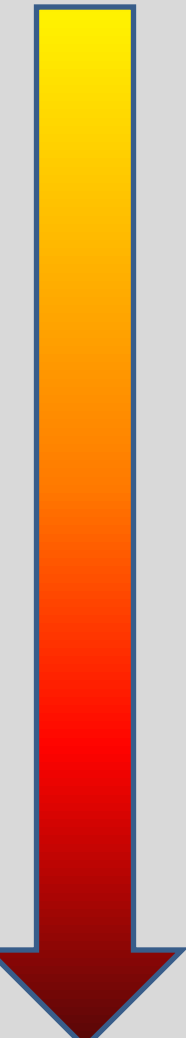
Therapeutic use:

→ Smoking cessation aid to reduce cravings and withdrawal symptoms

Pregnancy:

- The incidence of birth defects following nicotine replacement therapy may be similar to those of smoking.
- Behavior recommendations for smoking cessation are recommended.

Acute Nicotine Toxicity (concentration-dependent)

- 
- Cholinergic excess
 - Central stimulation
 - Hypertension, tachycardia
(sympathetic stimulation at
ganglia and adrenal medulla)
followed by
hypotension, bradycardia
(depolarizing block of
autonomic ganglia)
 - **Convulsions**
 - **Death by respiratory arrest**
central respiratory center depression
and respiratory muscle paralysis

Depolarizing block mechanism:

- Initial stimulation of central, autonomic, and neuromuscular nicotinic receptors
- Prolonged agonist occupation of receptors causes persistent depression



- **depolarizing blockade**

Blockade at motor end plate →

- **muscle fasciculations**
- **muscle weakness / flaccid paralysis**
- **respiratory muscle paralysis**

Varenicline pharmacology

PK: Oral; bioavailability ~90%
Low plasma protein binding
Minimal hepatic metabolism
Renal excretion of unchanged drug (OCT2 substrate) → renal dose adjustment
 $t_{1/2\text{elimination}}$ ~24 hours

Action: partial agonist of $\alpha 4\beta 2$ neuronal nAChR

stimulates dopamine activity of ventral tegmental area *plus* blocks nicotine from binding
↓ nicotine craving and withdrawal symptoms

Use: smoking cessation aid to help reduce nicotine cravings and withdrawal symptoms

Adverse effects:

Nausea (take after eating with full glass of water) CNS: headache, insomnia, abnormal dreams,
Hypersensitivity (rare reports): skin, angioedema impaired physical and mental abilities

Pregnancy: The incidence of adverse pregnancy outcomes following maternal use of varenicline is not greater than the pregnancy outcomes of pregnant patients who smoke.

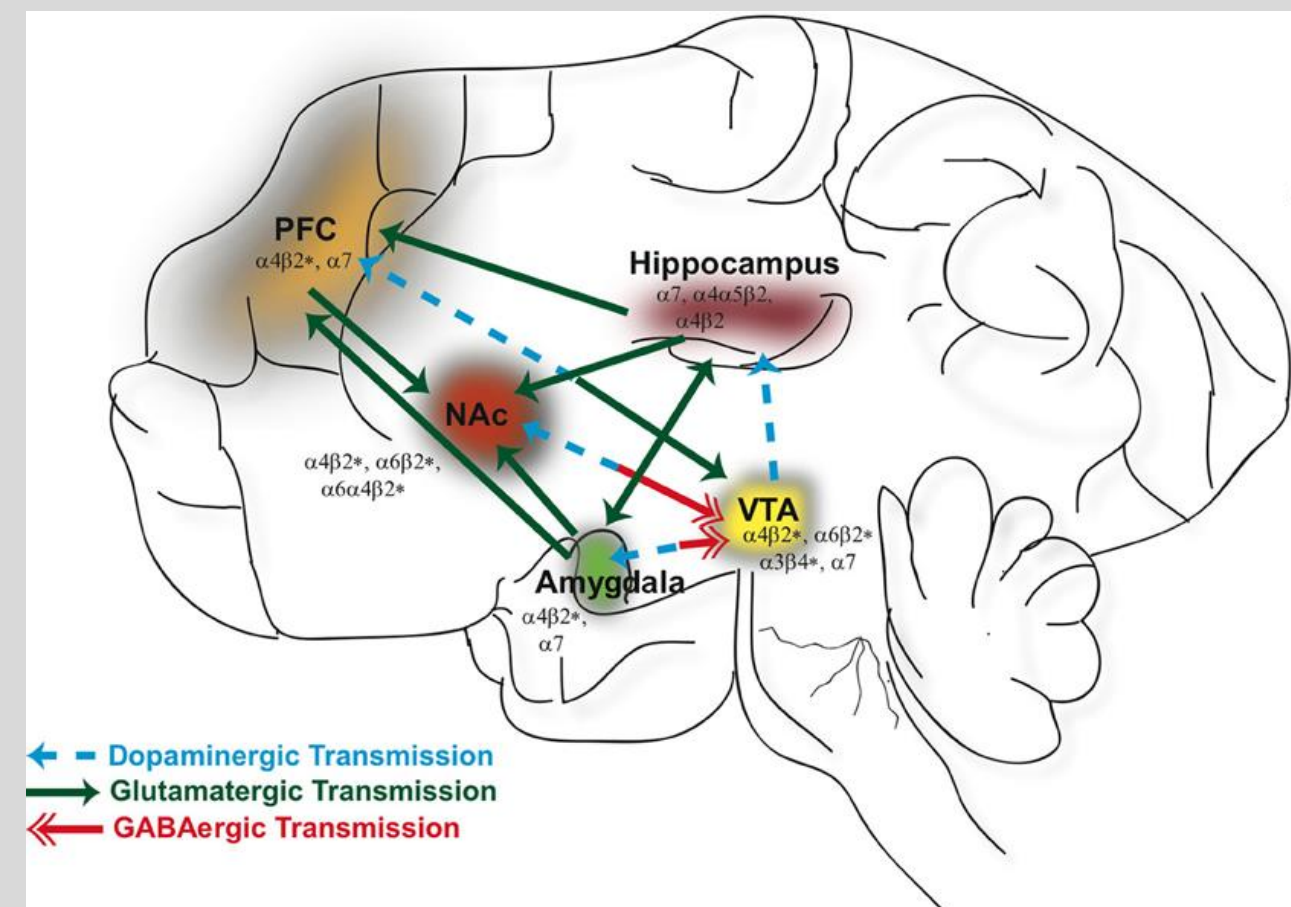
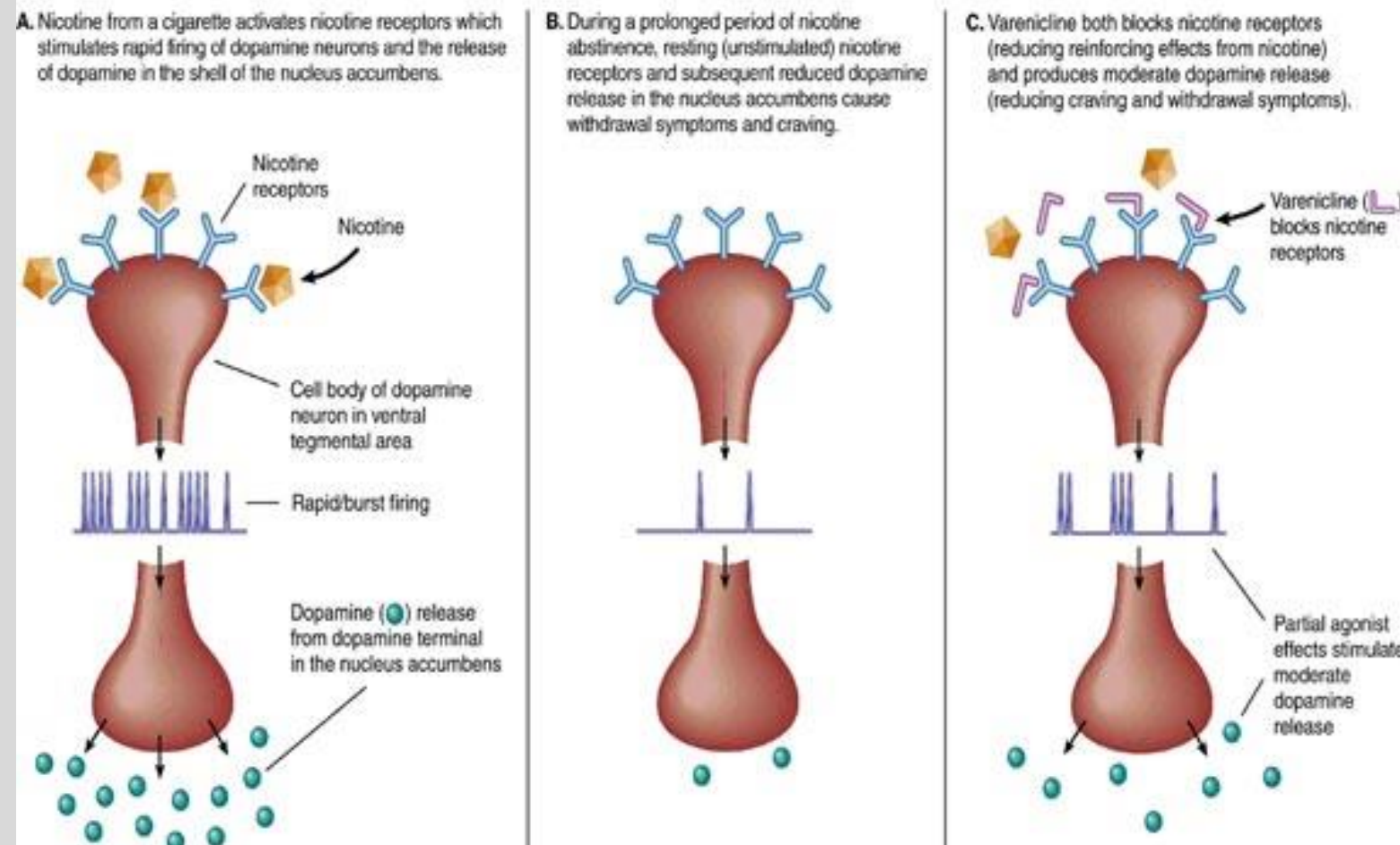
Drug interactions

Nicotine: Increased cholinergic effects

Alcohol: May enhance neuropsychiatric effects

Varenicline is partial agonist of $\alpha 4\beta 2$ neuronal nAChR

Figure 3. Highly simplified scheme showing effects of (a) nicotine from cigarettes (b) nicotine withdrawal and (c) varenicline, on nicotine receptors and dopamine release.



Foulds J. The neurobiological basis for partial agonist treatment of nicotine dependence: varenicline. International Journal of Clinical Practice 2006; 60:571-576.

http://www.treatobacco.net/en/page_184.php

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Allison A. Feduccia, Susmita Chatterjee and Selena E. Bartlett.
Neuronal nicotinic acetylcholine receptors: neuroplastic changes underlying alcohol and nicotine addictions. Front. Mol. Neurosci., 03 August 2012 | <https://doi.org/10.3389/fnmol.2012.00083>

Check your knowledge

- Where are nicotinic receptors located?
-
- What type of receptor are nicotinic receptors?
-
- A healthy adult patient wishes to quit smoking? What two drugs in this category can be recommended?
-
- What are the differences between these two drugs?
-
- What are the symptoms/signs of acute nicotine toxicity?
-

Summary

Summary: Direct-Acting Cholinomimetics

Acetylcholine intraocular	Miotic agent, adjunct ophthalmic surgery	• Rapid acting
Methacholine inhalation	Dx test: bronchial hyperreactivity	• Inhalation only • Severe bronchoconstriction • Medical supervision
Carbachol ophthalmic drops	Glaucoma Adjunct ophthalmic surgery	• Carbachol ophthalmic use only • Nicotinic + muscarinic properties when used systemically
Bethanechol Oral; variable abs	Stimulate urinary motility	• Generalized muscarinic effects • Lacks nicotinic actions
Pilocarpine ophthalmic drops oral	Angle-closure glaucoma Open-angle glaucoma Xerostomia (oral)	• Profuse sweating, salivation • CNS effects
Cevimeline oral	Xerostomia of Sjogren's syndrome or radiation damage of salivary glands	• M₃ affinity • Hepatic metabolism (CYP2D6,3A4)

Summary: Therapeutic Uses of Direct-Acting Nicotinic Receptor Agonists

Nicotine	Smoking cessation aid	• Depolarizing block • Poisoning may cause seizures, respiratory depression, death
Varenicline	Smoking cessation aid	• High affinity, selective $\alpha_4\beta_2$ nAChR partial agonist • CNS effects, nausea • Avoid alcohol and nicotine products

Rest and Digest



Tea at the Vicarage [Circa 1905] *Margaret Isabel Dovaston*

Check your knowledge

- What is the definition of cholinergic receptors?
- Acetylcholine is the neurotransmitter for nicotinic and muscarinic receptors.
- What are the cholinergic receptor subtypes?
- Nicotinic receptors are ligand-gated ionotropic receptors. Sodium, potassium, and sometimes calcium pass through them. Muscarinic receptors M_1 , M_2 , M_3 , M_4 , M_5 are G protein-coupled receptors located on effector cells. $M_{1,3,5}$ are postsynaptic GqPCRs. $M_{2,4}$ are GiPCRs, auto- and heteroreceptors.
- Where in the body are cholinergic receptors located?
- Nicotinic receptors in the periphery are located at all ANS ganglia and the adrenal medulla. The muscarinic receptors are distributed in the heart, blood vessels, bronchi, GI tract, urinary tract, and salivary, lacrimal, mucous, and sweat glands. Nicotinic and all types of muscarinic receptors are distributed across various regions of the CNS and non-neuronal cells, such as immune cells (N and M) and endothelial cells (M_3).

Check your knowledge

- Acetylcholinesterase can hydrolyze 5000 acetylcholine molecules per second. Why, then, is the duration of action of ACh instilled in the anterior chamber as long as 6 hours?
- In neuron synapses, AChE is highly concentrated, which is necessary for precise regulation of ACh activity, but there is a lower concentration of AChE in the anterior chamber of the eye→ reduced activity→ longer duration of ACh action.
- An asymptomatic 6-year-old patient suspected of having asthma has a positive bronchoprovocation test using an orally inhaled cholinergic agent. What are the mechanism of action and effects of the administered drug?
- Administration of methacholine by oral inhalation activates M3 receptors in bronchial smooth muscle, causing bronchoconstriction. In patients with pre-existing bronchial hyperreactivity, bronchoconstriction will be provoked at lower methacholine doses. A patient with a negative bronchoprovocation test is unlikely to have asthma. Bronchoprovocation testing may be useful for assessment of disease severity, prognosis, and plan of asthma treatment.
- What are the mechanism-related side effects of muscarinic receptor agonists?
- GI: sialorrhea, dyspepsia, abdominal upset, diarrhea, UT: urinary frequency, urgency, incontinence, CV: palpitations, Eye: blurred vision, visual disturbance, CNS: drowsiness, asthenia

Check your knowledge

- Where are nicotinic receptors located?
- CNS, ANS ganglia (both PNS and SNS), chromaffin cells of adrenal medulla
- What type of receptor are nicotinic receptors?
- Ligand-gated ionotropic ion channels activated by ACh binding to sites on α subunits adjacent to neighboring subunits. Opening of the channels increases permeability to Na^+ , K^+ , and in some channels Ca^{2+} .
- A healthy adult patient wishes to quit smoking? What two drugs in this category can be recommended?
- Nicotine, available in several different products, and oral varenicline.
- What are the differences between these two drugs?
- Nicotine is a full agonist. Varenicline is a partial agonist at $\alpha 4\beta 2$ neuronal nAChR. Both drugs help reduce cravings and withdrawal symptoms.
- What are the symptoms/signs of acute nicotine toxicity?
- Cholinergic excess (stimulation of receptors on effector cells), CNS stimulation sympathetic excess (effect at ganglia) followed by hypotension and bradycardia (depolarizing block of autonomic ganglia), convulsions, death by respiratory arrest

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- Access Medicine Goodman & Gilman's The Pharmacological Basis of Therapeutics, 14e, 2023; Chapter 11: Muscarinic Receptor Agonists and Antagonists
- Lexicomp drug monographs
- Lippincott® Illustrated Reviews: Pharmacology, 8e, 2023; Chapter 3: The Autonomic Nervous System

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